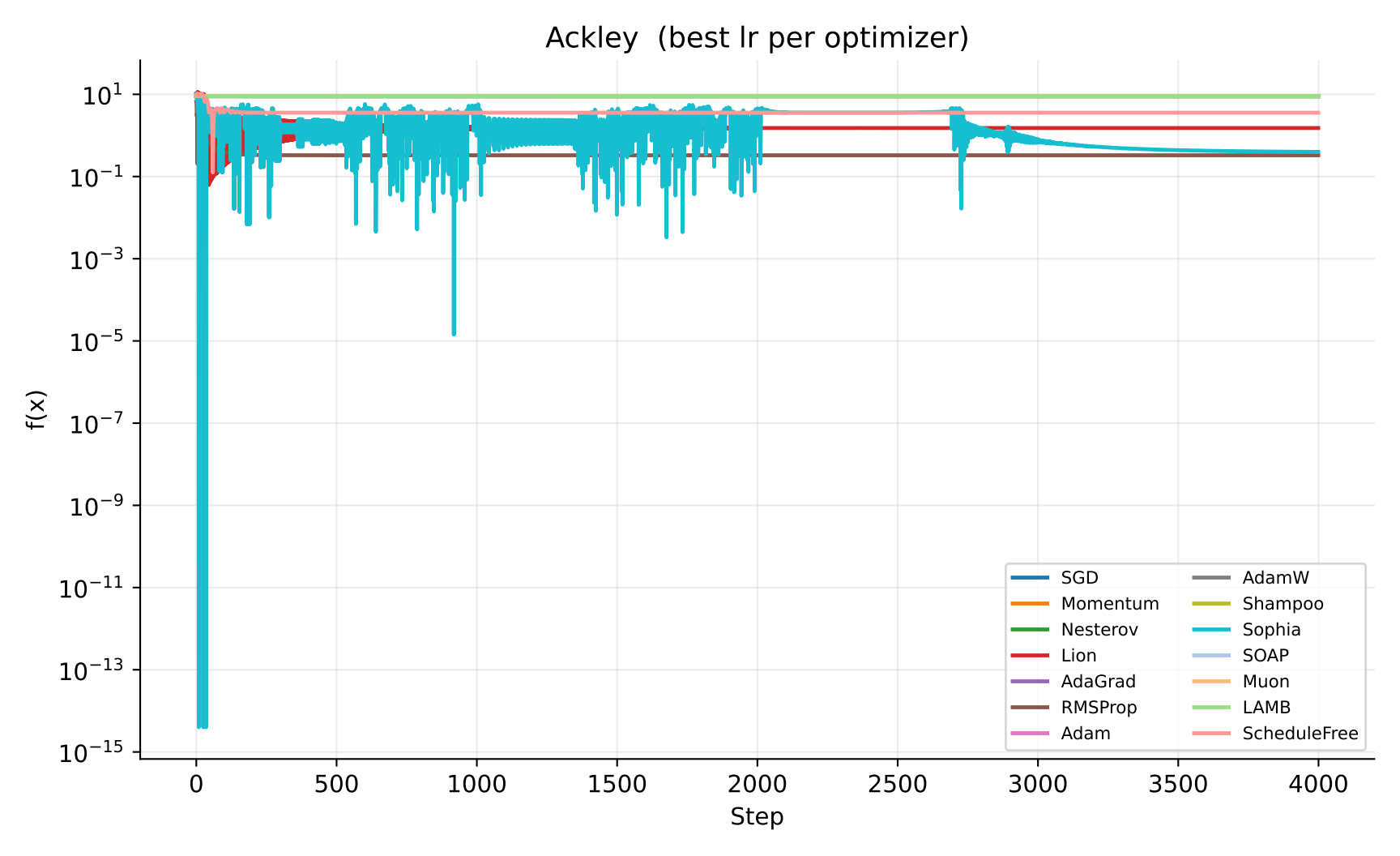

Ackley (best lr per optimizer)
f(x)
Step
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
0
500
1000
1500
2000
2500
3000
3500
4000
SGD
Momentum
Nesterov
Lion
AdaGrad
RMSProp
Adam
AdamW
Shampoo
Sophia
SOAP
Muon
LAMB
ScheduleFree